

CSE320 (Data Comm) – Practice Sheet on Chapter 10

Q1.

Suppose you want to transmit the message 11011011 and protect it from errors using the CRC generator polynomial $x^3 + 1$. Using binary division, show the message that should be transmitted.

Later, corrupt the left-most third bit of the transmitted message and show that the error is detected by the receiver using CRC technique.

Q2.

Assume a packet is made only of four 16-bit words $(3046)_{16}$, $(ABDC)_{16}$, $(2B5)_{16}$, and $(E30)_{16}$.

- I. **Show** the checksum at the sender.
- II. If the first data item is changed to $(3047)_{16}$ and the third data item is changed to $(2B4)_{16}$ during transmission, check if the receiver can detect any error in this case?
- III. Explain the reasons of the receiver's error detection state in (II)

(Hint: The given words are in hexa-decimal value, that means, each digit can be represented by 4 bits. Remember hexadecimal values range from 0000 – FFFF).

Q3.

The 2 bit datawords are converted to the following 5 bit codewords. For how many bits can we successfully detect and correct errors using this scheme?

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

Q4.

Suppose two channels are using the following linear block coding scheme for error detection and correction. In a linear block coding scheme, XORing two distinct valid codewords produces another distinct codeword.

Dataword	Codeword
00	00000
01	01011
10	10101
11	?????

- From the table above, find the value of n and k for the notation C(n,k).
- What would be the missing distinct codeword?
- Can we successfully detect and correct up to 2 bits error using the scheme?

Q5.

Assume the following cases for a sender and a receiver device that use CRC for error control.

Dataword: $x^6 + x^4 + x^3 + x + 1$

Divisor: $x^4 + x^2 + x + 1$

Show the sent codeword in CRC polynomial format.

Q6.

Suppose a sender is sending a message consisting of five 5-bit numbers to the receiver. In both ends, traditional checksum is being used to detect errors.

Sender		Receiver	
Data	8 12 14 0 9	Data	9 11 14 0 9
Actual Checksum	?	Received Checksum	19

- Is the actual checksum from the sender equal to the checksum received by the receiver?

- Are there any errors in the data received? **Explain** if the errors are detectable using checksum or not.

Q7.

Observe the Block Coding scheme below.

Dataword	Codeword
00	0001110
01	0110110
10	1010101
11	1100001

- Calculate the minimum hamming distance.
- How many errors can this scheme detect and how many can it correct?
- Receiver receives the two codewords 0110010 101110. Identify if there were any errors in the received codewords. For each codeword, find the original data after error correction?

Q8.

Suppose you want to crack promo codes for your favorite “BAAPJI” games. You know that the game publishers are using CRC to verify the legitimacy of an entered code. Upon some research, you partially cracked the following info for one of your entries.

- Codeword: $x^6 + x^5 + x$
- Quotient: $x^3 + x^2 + x$
- $C(7,4)$

However, sigh! No mention of the divisor. But a genius like you would never stop to crack your favorite game! So, you combined your current Data Comm. lesson and your high school math formula –

$$\text{Dividend} = \text{Divisor} * \text{Quotient} \oplus \text{Remainder}$$

Now show us how you did it by answering the following questions.

- From the given Codeword and $C(n,k)$, **find out** what the remainder was? Why so?
- What was the dividend? Why so?

- iii. From (i), (ii), and the formula, find out the correct divisor. **Show** full steps.

Hint: Binary and Polynomial CRC are interchangeable. You may use any of them, whichever feels suitable for a question. All operations are in modulo-2 arithmetic.

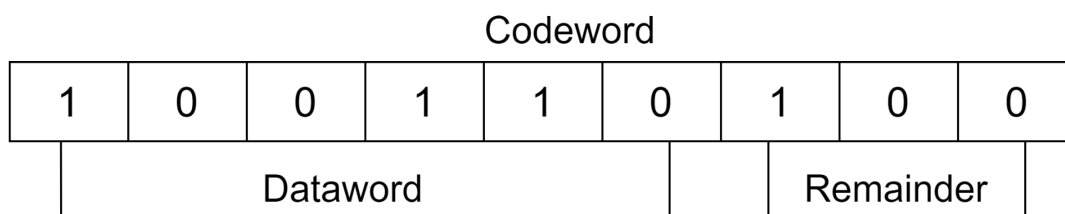
Q9.

Let us assume a packet is made only of four 16-bit words $(450F)_{16}$, $(1A4)_{16}$, $(A309)_{16}$, and $(B00)_{16}$. (Hint: The given words are in hexa-decimal value, that means, each digit can be represented by 4 bits. Remember hexadecimal values range from 0000 – FFFF).

- i. Show the checksum at the sender.
- ii. Synthesize a case where the checksum method will fail to detect error and show how the checksum method will fail. Explain the reason for failure.

Q10.

The following codeword is received by a receiver X. Receiver X uses CRC technology to detect errors. Determine whether there is any error in the received codeword or not. [The first 2 bits in the divisor are 1, rests are 0]



Q11.

Suppose we are using a (2,4) encoding scheme and the assignment of Dataword to Codewords are as follows:

Dataword	Codeword
00	0010
01	1001
10	1011
11	1100

- i. What is the minimum Hamming distance here?
- ii. How many bits can be corrected and detected?
- iii. The receiver receives 0111. Explain what type of error this is. Can this codeword be corrected? Justify your answer.